



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: **MCA AI&DS**
Semester: **II**
Name of the Paper: **Computer Network**
Paper Code: **TMD205**
Time: **1.5 hour**

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1 (10marks)

- Explain the concept of hosts and end systems in the context of computer networks.
(a) Provide a comprehensive list of different types of end systems (e.g., PCs, laptops, smartphones, servers) and discuss their roles in network communication. Is a Web server an end system? Justify your answer with appropriate reasoning.

**CO1,
CO2**

OR

- Compare and contrast the OSI Reference Model with the TCP/IP Protocol Stack. Create a detailed mapping table showing the correspondence between OSI layers and TCP/IP layers. For each layer in both models, identify the key protocols and functionalities. Discuss the advantages and limitations of each model in practical network design.

Q2 (10 marks)

Consider a network scenario where a data transmission occurs between two hosts connected by a single link with the following parameters:

- Link transmission rate: $R = 1,000$ Mbps
- Distance between hosts: $m = 5,000$ km
- (a) • Propagation speed: $s = 2.5 \times 10^8$ m/sec
- Packet size: $L = 2,500$ bytes

Answer the following:

- i. Calculate the transmission delay (d_{trans}) for the packet.
- ii. Determine the total end-to-end delay ($d_{end-to-end} = d_{trans} + d_{prop}$).

**CO2,
CO3**

OR

- Analyze the four components of end-to-end delay in packet transmission: processing delay, queuing delay, transmission delay, and propagation delay. For each component, explain:
- (b) i. What causes this delay?
ii. Is it constant or variable?
iii. How does it affect real-time applications such as video conferencing and VoIP?

Q3 (10 marks)

Explain the Hypertext Transfer Protocol (HTTP) with emphasis on persistent and non-persistent connections. For a web page containing one HTML file and four embedded images, calculate the total time required to retrieve the page using:

- (a) i. Non-persistent HTTP (assume $RTT = 50$ ms, server processing time = 10 ms)
ii. Persistent HTTP with pipelining (assume $RTT = 50$ ms, server processing time = 10 ms)



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OR

- (b) i. Discuss the role of cookies in HTTP communication. How do cookies maintain state information for stateless HTTP protocol? What are privacy concerns associated with cookies? CO3, CO4
- ii. Discuss the benefits of DNS caching and how it reduces query response times and network traffic.

Q4 (10 marks)

- Explain the role of proxy servers in web architecture and their benefits for reducing bandwidth consumption. Discuss conditional GET requests and the use of the "If-Modified-Since" header in HTTP. How does this mechanism reduce unnecessary data transmission? What are the implications for web server performance and client experience? CO3, CO4

OR

- (b) What is an application-layer message? A transport-layer segment? A network layer datagram? A link-layer frame? Which layers in the Internet protocol stack does a router process? Which layers does a link-layer switch process? Which layers does a host process?

Q5 (10 marks)

- Analyze the Email system architecture involving SMTP (Simple Mail Transfer Protocol) for sending messages and POP3/IMAP for retrieving messages. Answer the following:
- (a) i. Explain the complete sequence of operations when a user sends an email from one mail server to another using SMTP.
- ii. Compare POP3 and IMAP protocols in terms of email storage location, message management, and support for multiple devices. CO2, CO3

OR

- A media company plans to distribute a high-definition movie file to customers using different distribution architectures. Consider the following parameters:
- File size: $F = 8$ GB (gigabytes)
 - Server upload capacity: $u_s = 50$ Mbps
 - Number of customers (peers): $N = 25$
 - Minimum download rate per customer: $d_{\min} = 15$ Mbps
 - (b) - Average upload rate per customer: $u = 6$ Mbps
- i. For client-server distribution, calculate the minimum time D_{cs} required to distribute the file to all N customers. Consider both server upload constraint and client download constraint.
- ii. For P2P distribution (BitTorrent model), calculate the minimum distribution time D_{P2P} using the formula approach:
- $$D_{P2P} = \max \{ NF/u_s, F/d_{\min}, NF/(u_s + \sum u_i) \}$$